

Identifying the Greatest Tour de France Riders Through Network Science

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In this project, we construct a series of directed multi-digraphs to analyze over a century of Tour de France results (1903–2024) through means of network analysis. For each stage, a directed edge is created from every participant to all riders who placed ahead of them, forming a competitive hierarchy for that stage. By aggregating data from all stages across all editions of the race, we generated a massive multi-digraph capturing the cumulative performance dynamics of Tour de France participants. Each multidigraph differentiates in its weights, resulting in different scores after the PageRank algorithm is applied. With those results we are able to rank riders based on the structure of competitive dominance inherent in their race history. This approach offers a novel perspective on historical performance, highlighting riders who consistently placed ahead of strong competitors, regardless of raw wins or podium finishes.

Introduction Tour de France is the most prestigious cycling event in the world. The multi-stage competition tests the best cyclists across various terrains, from flat sprints to punishing climbs. While general classification highlights the best overall cyclists, it misses certain cyclists who excel in specific domains such as consistent mountain climbers, sprinters and other stage specialists.

This project aims to accumulate all the Tour de France history by constructing a network that captures the relative performance of all riders in each stage from 1903 to 2024. Historical results are obtained from ProCyclingStats.com, a comprehensive and publicly available cycling statistics database. For each stage, a directed edge is created from a rider to every other rider who placed ahead of them, forming a stage-level performance network. These individual networks of stages are then aggregated into a single multi-digraph, where nodes represent riders and multiple directed edges capture how frequently and consistently one rider was outperformed by another. To better reflect the different dimensions of performance, we construct several variants of this network using different edge weighting strategies. For example, weighting edges based on time gaps, positional gaps, or stage types (e.g., flat, hilly, mountain, time trial). This allows us to analyze the data from multiple perspectives and to highlight specific rider strengths that may not be apparent in overall classification rankings.

To rank the riders within these networks, we apply the PageRank algorithm (1), originally developed to rank web pages based on link structure. In our context, PageRank assesses not just how frequently a rider finishes ahead of others, but also the strength of the competitors they outperform. As a result, riders who consistently place ahead of highly ranked opponents receive higher scores. This method allows us to identify not only dominant overall champions, but also consistently strong performers and underrated specialists who might be overlooked by traditional rankings.

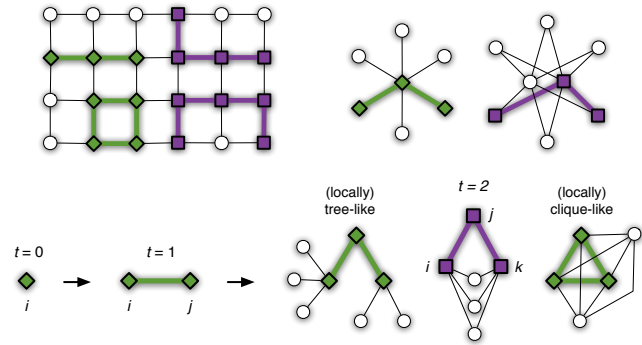


Fig. 1. Mandatory informative illustration highlighting main contributions. (?)

For the visualization and interpretation of the results, we first apply the Leiden clustering (2) algorithm to detect community structure in the rider network. This groups riders into a generation of riders. To visualize the network, we use the Fruchterman-Reingold (3) force-directed layout, which positions riders in a two-dimensional space based on their network proximity.

Related work

The application of network analysis in sports research is vast; it can be used to uncover team dynamics, player interactions and performance metrics across various sport disciplines and much more. For instance, Bi et al. (4) applied network analysis to examine how socio-demographic factors such as age, gender, and education influence sports participation in China. In team sports such as football and basketball, passes are a crucial aspect of the game. The works by Pena & Touchette (5) as well as Fewell et. al. (6) explore the importance of passing by constructing passing networks and analyzing their structural properties and uncovering unique team strategies.

The PageRank algorithm (1), originally developed to rank web pages, has been successfully adapted to the realm of sports analytics. Several researchers have demonstrated its utility as a robust tool for ranking players and predicting match outcomes. For example, Matthews et al. (7) employed a weighted PageRank approach for NCAA basketball, incorporating factors such as score margin, date, and venue, and demonstrated that it could outperform traditional seeding methods. On a different note, Mukherjee (8) used PageRank together with centrality measures to analyze the performance of cricket players, constructing a directed and weighted network based on

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individual match interactions. Shi and Tian (9) proposed a generalized PageRank model that integrates Bayesian correction with game outcomes to enhance the prediction accuracy of team rankings.

Beyond team sports, network analysis has also been effectively applied to individual sports. In tennis, Radicchi (10) applied PageRank to head-to-head match data to identify top players in history, demonstrating the algorithm's strength in individual, tournament-based sports. In Formula One, Horvat et al. (11) modeled driver performance using directed multi-graphs, where drivers are nodes and finishing positions are represented as edges. These networks allow for the application of centrality and community detection algorithms, including PageRank, to identify dominant drivers and race patterns across seasons. In this project, we adopt and extend Horvat's methodology to the Tour de France by considering different kind of constructed networks and testing the scalability of their approach on a much larger network.

In terms of cycling, no major studies delve into ranking the historic achievements of cyclists. However there are various studies that focus on the prediction of upcoming race winners presented by Kholikine et al. (12) and the prediction of upcoming talent for example in the work by (13) but none of those studies use network analysis approaches.

Results

Main results supported by math, plots, tables, diagrams etc. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Table 1. Table describing data or methods.

	n	m	$\langle k \rangle$	$\langle C \rangle$	$\langle d \rangle$
Fine network	438 920	9 742 733	44.4	0.37	6.19
Random graph	438 920	9 781 609	44.6	0.00	4.92

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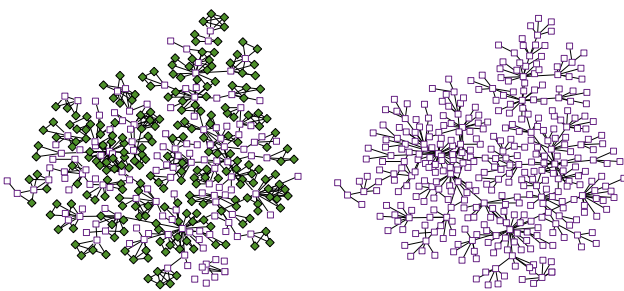


Fig. 2. Figure showing interesting examples. (?)

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Discussion

Summary of results, main contributions, final con-

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Methods

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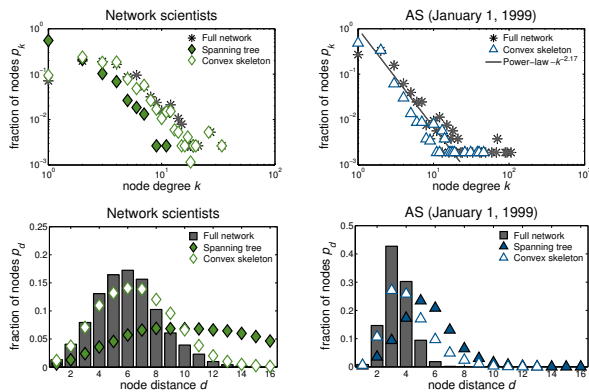


Fig. 3. Figure showing relevant results. (?)

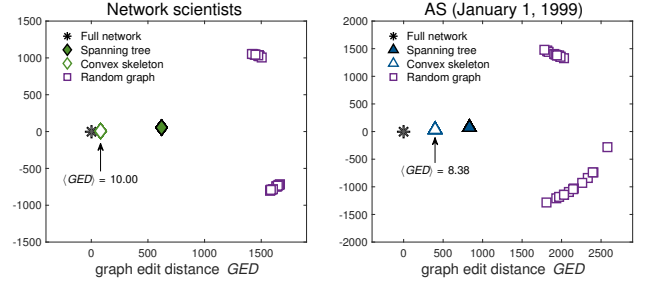


Fig. 4. Another figure with results. (?)

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Require: graph G , cutoff k_{min}

Ensure: power-law γ

- 1: $s \leftarrow n \leftarrow 0$
 - 2: **for** nodes $i \in N$ **do**
 - 3: **if** $k_i \geq k_{min}$ **then**
 - 4: $s \leftarrow s + \ln k_i / (k_{min} - 0.5)$
 - 5: $n \leftarrow n + 1$
 - 6: **return** $1 + ns^{-1}$
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